A Sigma-Delta Fractional-N Frequency Synthesizer based on ADPLL

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***Abstract*— in this paper, we propose a fractional-N frequency synthesizer based on All Digital Phase Locked Loop (ADPLL). We use phase-frequency detector as PD, up/down counter as LF, P-Divider counter as DCO, and the dual-modulus N/N+1 divider is controlled by the output of the sigma-delta modulation to achieve the goal of spur reduction. The Sigma Delta Modulator (SDM) have noise-shaping characteristic though SDM will introduce inevitable quantization noise. We discuss the phase noise caused by the quantization noise brought by the Sigma-Delta noise-shaping technique.**

Keywords: Frequency Synthesizer, ADPLL, Fractional-N, Sigma-Delta, Noise-Shaping

I. INTRODUCTION

Frequency synthesizer is a component which generates many frequencies from one reference frequency. It is widely used in the fast developing wireless communication area, and it is a critical component in the wireless communication system as a local oscillator for wireless channel selection.

Phase locked loop is used to design a frequency synthesizer with high clock accuracy. The traditional analog PLL is very sensitive to parameters, and must be recalculated if the application is changed. All Digital Phase Locked Loop is used to implement frequency synthesizer to overcome the shortcomings of his analog counterpart, ADPLL has the advantage of fully integrated, low-cost and low-power. ADPLL have become more attractive these years because it provides a faster lock-in time and better stability and portability over different process.

The fractional-N frequency synthesizer can achieve a small frequency resolution while a large reference is used. We change the loop divide ratio periodically between N and N+1 to realize fractional divider ratio, but this leads to an unwanted disadvantage of the low frequency spur. Sigma Delta noise shaping technique can be used in the fractional-N frequency synthesizer for spurs reduction.

The presented fractional-N ADPLL use the SDM to randomize the instantaneous loop divide ratio, so that we can eliminates the fractional spur to overcome the disadvantage of the conventional fractional-N frequency synthesizer. Sigma Delta Modulator has a theoretical disadvantage of quantization noise because there is a nonlinear quantizer in the modulator. We will discuss the phase noise caused by the quantization noise.

This paper presents a Sigma Delta fractional-N frequency synthesizer based on ADPLL which is depicted in Figure 1.

Section 2 describes the components in our ADPLL. ADPLL is very different from its analog counterpart. There are only logic parts in the system. The design of ADPLL complies with the common knowledge of PLL.

Section 3 describes the principle of SDM, and show the noise-shaping characteristic through the derivation.

Section 4 describes the SDM used in ADPLL, and describes the phase noise caused by quantization noise. Power spectral density of phase noise of our ADPLL is simulated.

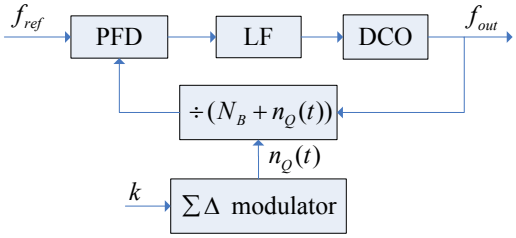


Figure 1. Diagram of proposed ADPLL FS

II. Components in the proposed All Digital Phase Locked Loop

ADPLL is a fully digital system which only contains logic components. The signal which transmitted in the ADPLL system can either be word signal or bit signal.

A. *Phase Detector (PD).*

The phase detector detects the phase difference between the reference and the feedback signal from the DCO or the frequency divider. Although the PD can be a XOR gate or a J-K flip-flop, we always use phase-frequency detector as our PD for a frequency synthesizer application. In our proposed ADPLL, we use memory elements flip-flop to implement the PFD component which is shown in Figure 2. We can realize that the two D flip-flop will be reset if both "UP" and "DOWN" are "1". The PFD functionality is depicted through its state machine in Figure 3.

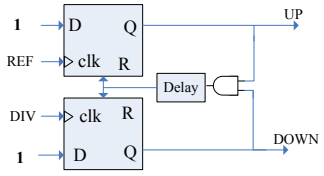


Figure 2. Diagram of PFD

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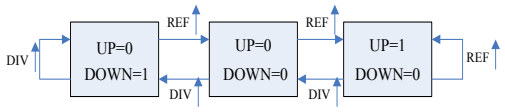


Figure 3. State machine of PFD

B. *Lowpass Filter (LF)*

In the proposed ADPLL, we use up/down counter as our LF which is depicted in Figure 4. The "Up/Down" and "CLK" are transformed from the "Up" and "Down" through pulse generator block. Here we should be aware of that the output of the up/down counter is word signal while the input is bit signal. The counter output is the weight of the input "UP" and "DOWN" pulse, so it can be approximately seen as an integrator.

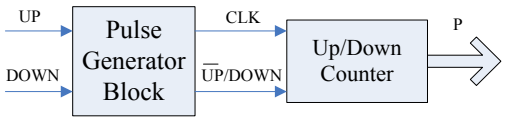


Figure 4. Diagram of LF

C. *Digital Control Oscillator (DCO)*

DCO, depicted in Figure 5, is based on a P-Divider Counter, with fixed high frequency as clock input, and the modulus P is controlled by the LF output.

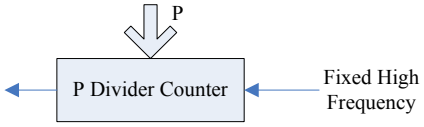


Figure 5. Diagram of DCO

D. *Dual-modulus N/N+1 divider*

The divider value (between N and N+1) is controlled by the output of the Sigma-Delta output. By varying the divider number between N and N+1 dynamically, a fractional divide ratio can be obtained.

III. The Principle of Sigma Delta Modulator

Figure 6 is the diagram of the first-order SDM. Q represents a quantizer which only has two levels. The linear model of Q is also depicted in Figure 6. The delay loop in the model is an integrator. At its output, Integrator cumulates the difference between the input and the feedback signal. The output level of Q will change when the output of the integrator is larger than the quantizer threshold value. Then the integrator input polarity is changed and the output of the integrator will change direction at next cycle. So that the feedback system forces the average of the modulator output equals the input signal.

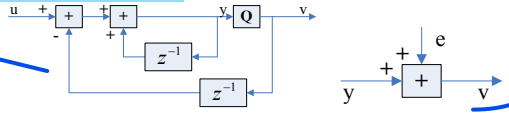


Figure 6. Diagram of First-Order SDM

From the diagram:

$$y(n) = y(n-1) + u(n) - v(n-1)$$
$$v(n) = y(n) + e(n)$$

So: $$V(z) = U(z) + (1 - z^{-1})E(z)$$

The noise transfer function:

$$|NTF(f)| = |1 - z^{-1}|_{z=e^{j2\pi f/f_s}} = 2\sin(\pi f / f_s)$$

In the proposed ADPLL we use a second order sigma delta modulator depicted in Figure 7, and also we can derive that

$$V(z) = U(z) + (1 - z^{-1})^2 E(z)$$

And the noise transfer function is

$$|NTF(f)| = |(1 - z^{-1})^2|_{z=e^{j2\pi f/f_s}} = (2\sin(\pi f / f_s))^2$$

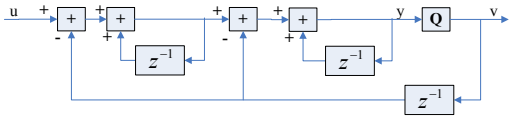


Figure 7. Diagram of Second-Order SDM

The second-order has more noise suppress than first-order modulator.

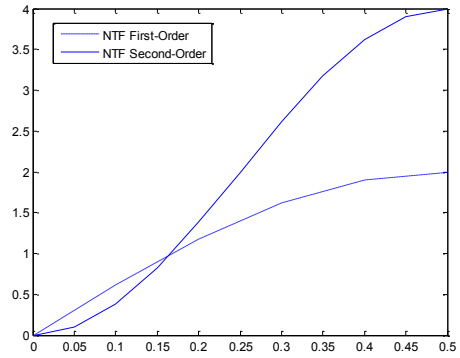


Figure 8. Noise Transfer Function

Figure 8 show the comparison between first order and second order SDM. We can see that 2nd SDM has more noise suppose in low frequency. The noise power is pushed to high frequency. This is so called noise-shaping characteristic.

IV. Sigma Delta Frequency Synthesizer

The Sigma Delta fractional-N ADPLL based frequency synthesizer compensates the fractional spur in the digital domain. The digital noise-shaping sigma delta modulator is used to randomize the instantaneous loop divide ratio. The frequency division ratio is controlled by the Sigma Delta output. The instantaneous division ratio is the sum of a base integer NB, and the integer output of the Sigma Delta Modulator n(t), so the average fractional division ratio in the feedback loop is $N = N_B + \overline{n(t)}$, where $\overline{n(t)}$ is the average output of SDM, and in our proposed ADPLL, the value of

$\overline{n(t)}$ is between 0 and 1. $\overline{n(t)} = \frac{k}{M}$, where k is the input number of the SDM, and M is the modulus used in the SDM. When the ADPLL reaches its steady state, its output frequency is:

$$f_{out} = N \cdot f_{ref} = \left[N_B + \overline{n(t)}\right] \cdot f_{ref},$$

And our ADPLL can reach the frequency resolution $\Delta f = \frac{1}{M} \cdot f_{ref}$.

When we use SDM in proposed ADPLL, we have two advantages. Randomizing the instantaneous division ratio can suppress fractional spur, while noise-shaping characteristic can pushes the phase noise associated with the divider from low frequency to high frequency, and the loop filter filters out the phase noise in high frequency.

Though there are advantages in Sigma Delta ADPLL, we should considerate phase noise caused by quantization noise. Open-Loop approximation is used to estimate the PLL output phase noise. This approach opens the connection between DCO and frequency divider and assume the input to the divider is an ideal signal with the desired frequency $N \cdot f_{ref}$, so the phase noise generated by the frequency divider is:

$$S_{\theta div}(f) = \left[\frac{f_{ref} \cdot Q(f)}{f \cdot N}\right]$$

where $Q(f)$ is the RMS spectral density of the quantization noise.

$$Q(f) = \sqrt{\frac{1}{12 f_{ref}}} \left|H_{NTF}(z)\right|_{z=e^{j2\pi f}}$$

We view $S_{\theta div}(f)$ as an equivalent input phase noise and use a closed-loop input to output phase transfer function to estimate output phase noise generated by the SDM.

$$S_{\theta out}(f) = S_{\theta div}(f)\left|H_{cl}(f)\right|^2$$

and $H_{cl}(f) = N$ within the PLL bandwidth. So in our case we can derive that:

$$S_{\theta out}(f) = \frac{f_{ref}}{12 f^2}\left[2\sin\left(\frac{\pi f}{f_{ref}}\right)\right]^4$$

Figure 9 is the MATLAB simulation result about our proposed frequency synthesizer.

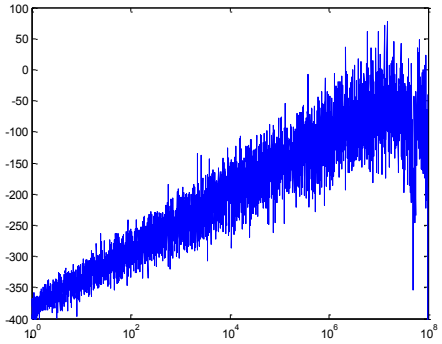


Figure 9. PSD of the phase noise

In the Figure, we can see that most fractional-N spurs have been suppressed.

For better performance, we should use higher order Sigma Delta Modulator. Higher order will introduce more instability and more complexity. We must design carefully if we choose higher order modulator.

V. CONCLUSION

This paper presents a ADPLL based Fractional-N Frequency Synthesizer which uses Sigma Delta noise-shaping characteristic to suppress the fractional spur. This ADPLL can be quickly deployed on Field Programmable Gate Array by HDL language and we should considerate the phase noise caused by SDM quantization noise.

REFERENCES

[1] R.M.Gray, "Oversampled sigma-delta modulation", IEEE Transactions on Communications, vol. COM-35, pp.481-489, Apr. 1987.

[2] R.M.Gray, "Spectral analysis of quantization noise in a single-loop sigma-delta modulator with dc input", IEEE Transactions on Communications, vol. 37, no. 6, pp.588-599, June 1989.

[3] Keliu Shu, CMOS PLL Synthesizers: Analysis and Design, Science Press, 2007, p69-73

[4] Michael H.Perrott, "A Modeling Approach for Sigma-Delta Fractional-N Frequency Synthesizers Allowing Straightforward Noise Analysis", IEEE Journal of Solid-State Circuits, Vol.37, No.8, 2002.4

[5] Schreier R. Temes G.C. "Delta-Sigma Data Converters", Science Press, 2007.1